

Research

Performance and improvement of deep learning algorithms based on LSTM in traffic flow prediction

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Abstract

Existing traffic flow prediction research lacks adaptability to complex traffic scenarios and has limited prediction accuracy. This paper introduces an improved LSTM (Long Short-Term Memory) algorithm and sliding window technology to improve the accuracy and stability of traffic flow prediction. This article discussed the enhancement of traffic flow (TF) prediction using a hybrid LSTM-BiGRU-Attention model aimed at improving the accuracy of traditional LSTM models. By integrating BiGRU (Bidirectional Gated Recurrent Unit) and Attention mechanisms, the proposed model captured complex TF patterns and long-term dependencies more effectively. The hybrid model is applied to Beijing urban road data with a time granularity (TG) of 10 min and a window size of 30 min, achieving an RMSE (root mean square error) of 4.478, an MAE (mean absolute error) of 3.609, and an R^2 of 0.965. The LSTM-BiGRU-Attention model had similar prediction errors in TF during peak and off-peak periods and maintained high stability under different TF conditions. LSTM-BiGRU-Attention outperformed a single LSTM model and other benchmark models in multiple performance metrics. When the TG is 10 min, the window size of 30 min has the best performance, with 4.478, 3.609, and 0.965, respectively. The combination of LSTM, BiGRU, and Attention provides an effective solution for the development of intelligent transportation systems, which helps to achieve more accurate TF prediction and optimized management.

Highlights

- (1) This model combines LSTM, BiGRU and Attention mechanisms to improve the accuracy of traffic flow prediction, especially when dealing with long-term dependencies and nonlinear features.
- (2) The data covers indicators such as traffic volume, speed and lane occupancy, and resamples at different time granularities.
- (3) This paper optimizes the input of time series data by setting different window sizes, and finally selects appropriate sliding windows and time granularity to improve the performance of the prediction model.

Keywords Traffic flow prediction · City road · Long short-term memory · Bidirectional gated recurrent unit · Hybrid model · LSTM-BiGRU-attention model

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1 Introduction

With the acceleration of urbanization and the rapid growth of the number of motor vehicles, traffic congestion has become increasingly serious [1, 2], becoming one of the important factors affecting the quality of life of urban residents and urban economic development. Accurate time flow prediction is of great significance for alleviating traffic congestion, optimizing traffic management and planning, and improving the efficiency of the transportation system. In intelligent transportation systems [3, 4], time flow prediction is a key component, that can provide decision support for traffic management departments by predicting future traffic conditions, such as signal control, path planning, congestion warning, etc. Time flow prediction using urban road data can provide a scientific basis for highway management and planning [5, 6]. With the improvement of big data technology and computing power, deep learning methods have shown great potential in dealing with complex nonlinear problems, especially in time series prediction [7, 8]. Long short-term memory networks [9, 10] are widely used in TF prediction because they can effectively capture long-term dependencies [11, 12]. By designing and optimizing deep learning models, the accuracy and stability of TF prediction can be improved.

To improve the accuracy and stability of TF prediction, this paper combines BiGRU and LSTM-based Attention methods. Beijing urban road data is collected, and then the data is preprocessed and resampled. In the study, the LSTM-BiGRU-Attention model outperformed the other models, with the lowest RMSE and MAE and the largest R^2 . Combining LSTM for long-term dependencies, BiGRU for bidirectional information, and Attention for dynamic weight assignment, the model showed consistent performance during peak and off-peak hours. Despite the high computational demands, this approach improved TF prediction accuracy and supported traffic management decisions.

The research in this paper has made significant contributions traffic flow prediction, which is mainly reflected in the following aspects. First of all, this paper introduces the LSTM-BiGRU-Attention model and applies deep-learning technology to traffic flow prediction, breaking through the limitations of traditional time series prediction methods. LSTM is good at capturing long-term dependencies in time series, while BiGRU improves the model's capturing capabilities through bidirectional information processing. The Attention mechanism further enhances the ability to pay attention to important time steps, thereby improving the model's prediction accuracy in complex traffic flow data. Secondly, this paper systematically explores the impact of window size on prediction performance at different time granularities and experimentally verifies the importance of selecting the optimal window size for optimizing prediction results at 10-min, 15-min, and 30-min granularity. This finding provides the theoretical basis and practical guidance for parameter selection of future traffic flow prediction models and helps to improve the decision support capabilities of traffic management systems. Third, this paper provides empirical data support for traffic flow prediction at different time scales by carefully resampling and analyzing the traffic flow data during the National Day Golden Week in Beijing. The study found that different time granularities and window sizes can effectively capture different traffic trends, revealing the complex dynamic characteristics in traffic flow prediction. These contributions not only enrich the theoretical system of traffic flow prediction but also provide new perspectives and methods for data-driven decision-making in actual traffic management.

The innovation of this article is mainly reflected in the following aspects. First, this paper combines LSTM with BiGRU and Attention mechanisms to propose a new deep-learning model to improve the accuracy of traffic flow prediction. The LSTM model excels at handling long-term time dependencies, while BiGRU enhances time series understanding through bidirectional information processing. On this basis, the Attention mechanism further improves the model's attention to important information through dynamic weight allocation. This innovative combination has not been widely used in traffic flow prediction, providing new ideas for research in this field. Secondly, this paper systematically evaluates the impact of window size on prediction performance under different time granularities, and through detailed experimental analysis, reveals the best match between short-term and long-term windows. This innovation not only optimizes the performance of the model but also provides a scientific basis for the selection of model parameters. The proposal of this method fills the gap in traffic flow prediction research and has high practical value. Finally, this paper conducts a detailed analysis of actual data during the National Day Golden Week, demonstrating the practical application effect of traffic flow prediction during peak periods. This kind of research based on actual data not only verifies the effectiveness of the model but also provides valuable experience for future applications in similar high-traffic scenarios. In short, this paper has made important contributions to model innovation and practical application and promoted the development of traffic flow prediction research.

The main contribution of this study is to systematically explore the time series data analysis methods based on the LSTM model, especially in the field of traffic flow prediction. This paper proposes an innovative sliding window strategy to optimize the selection of input data, thereby improving the model's ability to capture long-term dependencies

and periodic changes. Secondly, by comparing the effects of different resampling techniques, a set of effective data preprocessing standards are established to address the noise and missing value problems in time series data. In addition, this study also introduces advanced hyperparameter optimization methods to improve the training efficiency and prediction accuracy of the LSTM model, ensuring the reliability and practicality of the results. The innovation of the LSTM-BiGRU-Attention hybrid model in this paper lies in the integration of LSTM, BiGRU and Attention mechanisms, making full use of the advantages of each model and enhancing the attention to long-term dependencies, contextual information and important time steps. This combination improves the model's ability to capture the nonlinear characteristics of traffic flow data, significantly improves prediction accuracy, and overcomes the limitations of a single model in processing complex data. Finally, based on experimental verification of actual traffic flow data, the effectiveness in complex environments is demonstrated, providing empirical support for future intelligent traffic management systems. These contributions not only enrich the theoretical framework of time series analysis but also provide important references for application research in related fields.

2 Related work

In intelligent transportation systems, TF prediction has drawn a lot of interest. Machine learning [13] techniques have shown promise in this regard, using large amounts of historical data to find patterns and trends that increase prediction accuracy and dependability [14, 15]. Although support vector machines are accurate and have strong generalization capabilities, they have difficulty processing huge amounts of data computationally [16]. By merging predictions from decision trees and random forests, an ensemble learning technique, the stability and accuracy can be improved [17]. Because of its effectiveness and interpretability, random forests are frequently used in TF prediction to reduce overfitting problems through unpredictability [18]. Although the k-nearest neighbor approach is easy to use and straightforward, it becomes computationally costly and highly susceptible to parameter selection when dealing with high-dimensional and large-scale data [19]. Despite significant progress in TF prediction, there are challenges in achieving high performance during application.

A focus of the study is long short-term memory networks, which are a particular kind of recursive neural network that does well in TF prediction. Numerous research [20, 21] have shown that LSTMs are promising for TF prediction, speed forecasting, and travel time estimation. Du Bowen et al. [22], for example, provided an LSTM-based urban road TF prediction model and demonstrated its accuracy across a range of periods. The temporal dynamics and periodicity of TF data were well captured by LSTMs [23, 24], which also provided flexibility in unexpected event prediction scenarios. For practical implementation, LSTMs require improvements in computational efficiency, data quality reliability, and spatial information use, despite their capabilities in TF prediction.

Researchers have proposed methods to improve LSTM performance in time series forecasting. Integrating the Attention mechanism assigns weights to important sequence parts [25]. Zheng Haifeng et al. combined the Attention mechanism with LSTM, significantly enhancing accuracy and stability [26]. Combining BiLSTM with other structures captures both forward and backward information, improving temporal dependency understanding [27]. Hybrid methods, like integrating residual self-attention with convolutional GRU, improve robustness and accuracy [28]. Challenges include integrating models effectively, avoiding overfitting, and enhancing TF prediction further. The innovation of this paper is to introduce an improved LSTM algorithm combined with sliding window technology to optimize the accuracy and efficiency of traffic flow prediction. In addition, by adjusting hyperparameters and improving model structure, the prediction performance in complex traffic environments is improved.

3 Methods for TF prediction

3.1 Data collection

The data was collected from major roads in Beijing from October 1 to October 7, 2022, covering the National Day Golden Week [29, 30]. To meet the time series analysis requirements of the model, the original data needs to be resampled. This paper resamples time granularities of 10, 15, and 30 min to conduct model training and prediction at different time scales.

Covers traffic flow information on major traffic arteries in Beijing. Data is collected through a variety of collection devices, including traffic monitoring cameras, road sensors, GPS tracking devices, etc. These devices monitor and record a large number of key traffic indicators, such as vehicle speed, traffic volume, lane occupancy, and vehicle distance. To ensure the quality and accuracy of the data, this paper first preprocesses the original data comprehensively, including removing outliers, noise processing, and data completion to ensure that the data in each period are continuous and representative. To meet the needs of the model for time series analysis, this paper performs multiple resampling operations on the original data to generate data sets with time granularity of 10, 15, and 30 min respectively. This resampling method enables us to analyze the changing trends of traffic flow at different time intervals and provide more diverse and applicable time series data sets for subsequent model training and prediction. In addition, this paper subdivides the traffic peak period and non-peak period to further explore the performance of the model under different traffic conditions.

10-min granularity: which is one time step every 10 min, suitable for fine-grained analysis that requires high time resolution, and can capture rapid changes in TF in a short period.

15-min granularity: which is one time step every 15 min, suitable for medium-time resolution analysis, able to balance data volume and time accuracy, and suitable for general TF prediction.

30-min granularity: which is one time step every 30 min, and suitable for forecasting over longer time spans, providing trend analysis and macro forecasting over a longer period.

The total time for collecting data is 10,080 min. After resampling at TG of 10, 15, and 30 min, the number of data is 1008, 672, and 336, respectively.

The data collection for this study mainly relies on a variety of monitoring equipment, including traffic monitoring cameras, road sensors and GPS tracking devices. Traffic cameras capture traffic flow, speed and lane occupancy by taking real-time photos of road conditions; road sensors are used to detect the frequency, speed and distance of vehicles passing by; GPS tracking devices collect vehicle driving trajectories. During the data collection process, all equipment works synchronously to ensure that high-quality traffic indicator data is obtained and uploaded to the data processing center in real time through the transmission system.

The 10-min resampling data is shown in Table 1.

The 15-min resampling data is shown in Table 2.

The 30-min resampling data is shown in Table 3.

To ensure data quality and validity, missing and outliers in the data are handled. For missing values, the mean is used to fill in, and for outliers, they need to be corrected or removed according to the actual situation.

When resampling time series data, the original data is resampled from the minute level to the hour level. The downsampling method is used to aggregate the data by calculating the average value per hour. At the same time, the missing values that may appear are handled, and the gaps during the upsampling period are filled using linear interpolation. Finally, the resampled data is verified to meet the expected timestamp alignment and data consistency to ensure the reliability of the analysis results.

The formula for filling in the mean is:

Table 1 10 min resampling data

Timestamp	Number of vehicles	Average speed (km/h)	Lane occupancy
2022/10/1 0:00	35	90.5	0.12
2022/10/1 0:10	38	85.7	0.16
2022/10/1 0:20	37	88.3	0.15
2022/10/1 0:30	36	92.1	0.13
2022/10/1 0:40	37	87.6	0.15
...	...	...	...
2022/10/7 23:10	39	82.9	0.18
2022/10/7 23:20	41	88.4	0.20
2022/10/7 23:30	42	86.2	0.18
2022/10/7 23:40	44	89.7	0.17
2022/10/7 23:50	40	87.1	0.18

Table 2 15-min resampling data

Timestamp	Number of vehicles	Average speed (km/h)	Lane occupancy
2022/10/1 0:00	36	87.6	0.24
2022/10/1 0:15	39	90.2	0.18
2022/10/1 0:30	40	89.8	0.22
2022/10/1 0:45	42	86.7	0.23
2022/10/1 1:00	40	88.3	0.26
...	...	...	...
2022/10/7 22:45	39	85.9	0.28
2022/10/7 23:00	42	88.6	0.29
2022/10/7 23:15	43	89.1	0.27
2022/10/7 23:30	44	87.3	0.27
2022/10/7 23:45	41	87.6	0.26

Table 3 30-min resampling data

Timestamp	Number of vehicles	Average speed (km/h)	Lane occupancy
2022/10/1 0:00	38	88.1	0.32
2022/10/1 0:30	39	91.4	0.37
2022/10/1 1:00	41	87.2	0.39
2022/10/1 1:30	43	90.5	0.32
2022/10/1 2:00	44	89	0.34
...	...	...	...
2022/10/7 21:30	45	86.5	0.44
2022/10/7 22:00	46	89.2	0.43
2022/10/7 22:30	43	87.6	0.42
2022/10/7 23:00	44	88.1	0.42
2022/10/7 23:30	42	91.4	0.39

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

Identify outliers through statistical methods, using box plots to detect data points outside the normal range. Data exceeding the mean ± 3 times the standard deviation are considered outliers. For outliers, they are removed. Estimation of missing values uses mean filling, and the mean of the data is used to replace the missing data.

3.2 Sliding window settings

Sliding window technology [31, 32] is a popular time series data preparation technique in TF prediction. Continuous time series data may be split into fixed-length subsequences for model input and training by sliding the window. Nonetheless, the model's performance is significantly impacted by the sliding window's size. To find the ideal window length, this article uses sliding windows to generate samples. Various window sizes are chosen for comparison analysis. Sliding window technology is widely used in time series prediction, signal processing, computer vision, network traffic monitoring, and other fields. By setting a fixed-size window and gradually sliding the window to extract local features, the amount of calculation is effectively reduced and the efficiency of the algorithm is improved.

Choosing an appropriate sliding window size is crucial when using time series data as model input. The window size determines the number of time steps in each subsequence, impacting the model's ability to capture temporal patterns. A

balance is essential: a small window may miss important periodic and trend information, while a large window can apply noise by including irrelevant data.

The number of samples under different windows is shown in Table 4.

3.3 LSTM-BiGRU-attention hybrid model

The structure of the LSTM-BiGRU-Attention [33, 34] hybrid model is shown in Fig. 1.

The LSTM layer in a model processes input time series data, capturing long-term dependencies via its gating structure. It extracts trends and periodic features initially. The Attention mechanism dynamically adjusts time step weights and prioritizes important steps for predictions. LSTM effectively solves the gradient vanishing problem of traditional RNNs when dealing with long-term dependencies by introducing a gating mechanism. It can memorize important information in long time series while suppressing the influence of irrelevant information. In addition, LSTM excels in capturing trends and cyclical changes and is suitable for time series prediction and dynamic data modeling.

For each time step t , the activation value of forget gate f_t is calculated to determine how much cell state C_{t-1} from the previous time step needs to be retained.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

W_f represents the weight of the forget gate, and h_{t-1} is the hidden state at the previous moment.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3)$$

i_t represents the input gate output and W_i represents the input gate weight.

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (4)$$

$\tilde{C}_t$ represents the candidate cell state at the current moment, and W_C represents the weight matrix of the candidate cell state.

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (5)$$

C_t represents the cell state at the current moment.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

W_o represents the output gate weight.

$$h_t = o_t \cdot \tanh(C_t) \quad (7)$$

h_t represents the hidden state at the current moment.

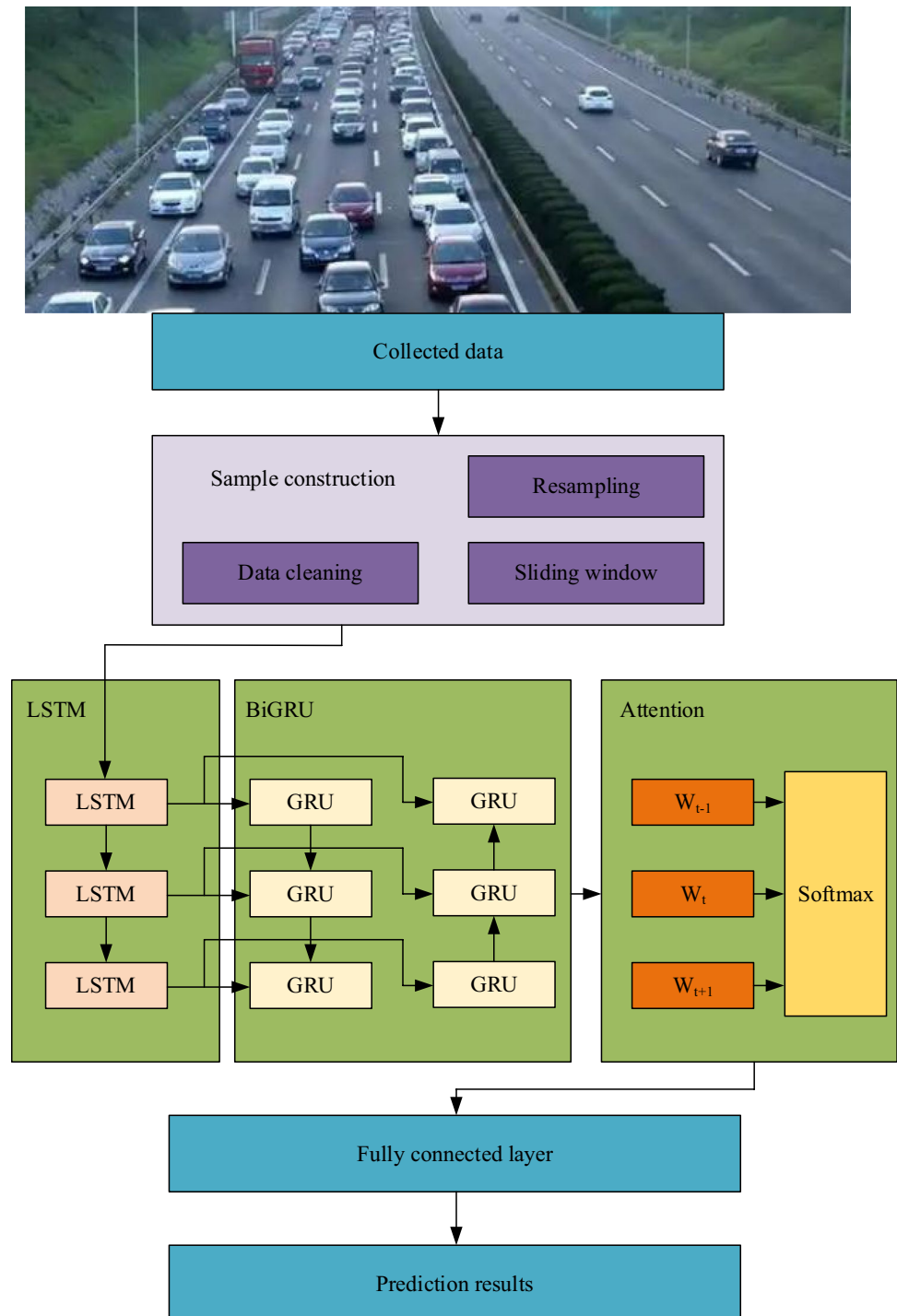
BiGRU [35, 36] is an extended version of GRU that can simultaneously process forward and backward time series data, thereby more comprehensively capturing features in time series.

The forward GRU is represented as:

$$\vec{h}_t = GRU(x_t, \vec{h}_{t-1}) \quad (8)$$

Table 4 Sample size under different windows

Window size (min)	Time granularity: 10 min	Time granularity: 15 min	Time granularity: 30 min
30	3	2	1
60	6	4	2
120	12	8	4
240	24	16	8

Fig. 1 LSTM-BiGRU-Attention hybrid model

The backward GRU is represented as:

$$\overrightarrow{h}_t = GRU(x_t, \overrightarrow{h}_{t+1}) \quad (9)$$

The bidirectional hidden state is represented as:

$$h_t = [\overrightarrow{h}_t; \overleftarrow{h}_t] \quad (10)$$

The attention score is calculated as follows:

$$e_{ij} = Q_i \cdot K_j^T \quad (11)$$

The attention score is normalized:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_k \exp(e_{ik})} \quad (12)$$

All value vectors are weighted and summed to obtain the output of the Attention mechanism. The application of Attention can dynamically adjust the weight of each time step so that when predicting TF, more attention is paid to the important time step information for the prediction results.

3.4 Model training and evaluation

The mean square error is used as the loss function, used to measure the difference between the predicted value of the model and the true value. The formula is expressed as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (13)$$

The training set is input into the model and the sliding window technique is used to generate fixed-length time series subsequences. The predicted values of the model are calculated through the LSTM layer, BiGRU layer, and Attention mechanism. Mean squared error is used to calculate the difference between predicted values and true values. The backpropagation is performed through the Adam optimizer to update the weight parameters of the model. The training steps are repeated until the loss function of the model converges on the training set.

To ensure the speed of model training, the experiment uses an Intel Xeon E5-2680 v4 2.40 GHz and an NVIDIA Tesla V100 graphics card. Using 128 GB of running memory and 2 TB of storage space, the operating system used is Ubuntu 20.04 LTS. The programming language used in the experiment is Python, and TensorFlow 2.6 is applied to build the LSTM-BiGRU-Attention model. NumPy 1.19 is used for numerical calculations and array operations. Pandas 1.3 is used for data processing and analysis. Scikit-learn 0.24 is used for data preprocessing and model evaluation.

The LSTM-BiGRU-Attention model is used to predict and analyze TF for windows at different time granularities, in order to determine the optimal window at different time granularities.

The output of the LSTM-BiGRU-Attention model is the predicted TF, which is the number of vehicles. To reflect the performance of TF prediction, RMSE, MAE, and determination coefficient R^2 are used for evaluation.

The formulas for evaluating indicators are:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (14)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (15)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (16)$$

To highlight the performance of the LSTM-BiGRU-Attention model more significantly, multiple models are compared. The comparison model set includes SVM (Support Vector Machine), ARIMA (Autoregressive Integrated Moving Average), LSTM, BiGRU, BiLSTM, LSTM-BiGRU, LSTM-Attention, and BiGRU-Attention.

4 Results

4.1 Prediction performance under different windows

LSTM-BiGRU-Attention is used for TF prediction, and the predictive performance under different windows is shown in Table 5.

Table 5 shows the predictive performance of different windows at three-time granularities, with window sizes of 30, 60, 120, and 240 min, respectively. When the TG is 10 min, the performance with a window size of 30 min is the best, with RMSE, MAE, and R^2 of 4.478, 3.609, and 0.965, respectively. When the TG is 15 min, the performance with a window size of 60 min is the best, with RMSE, MAE, and R^2 of 6.342, 5.234, and 0.946, respectively. When the TG is 30 min, the performance with a window size of 240 min is the best, with RMSE, MAE, and R^2 of 7.678, 6.435, and 0.938, respectively.

The reason for the different optimal window sizes at different time granularities is the need to balance the ability to capture short-term dynamic changes and long-term trends. When the TG is 10 min, a 30-min window size captures short-term fluctuations, improving prediction precision. For a 15-min granularity, a 60-min window balances short-term changes and medium-term trends, enhancing stability and accuracy. Table 5 shows the impact of window size on forecasting performance at different time granularities. At 10-min granularity, the 30-min window is optimal because it is more suitable for capturing short-term fluctuations. At 15-min granularity, the 60-min window is optimal, balancing short-term changes and medium-term trends. At 30-min granularity, the 240-min window is optimal, effectively capturing long-term trends. Different time granularities require different window sizes to optimize the balance between short-term dynamics and long-term trends and ensure the accuracy and stability of forecasts.

4.2 Prediction results at different time granularities

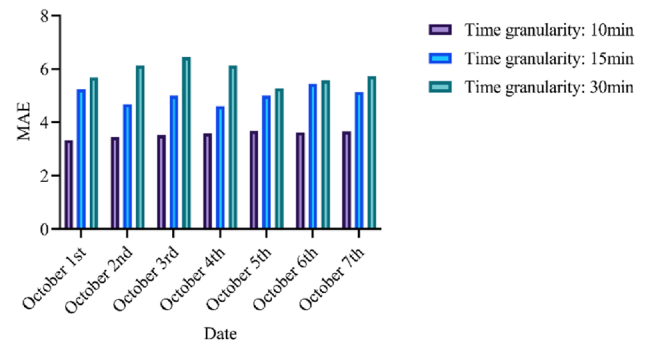
From Table 5, the optimal windows at three time granularities are selected. The TF prediction results at different time granularities are shown in Fig. 2.

The horizontal axis represents 7 days, and the vertical axis represents the MAE indicator. It can be seen that the MAE value of TF prediction with a TG of 10 min is the smallest, followed by a TG of 15 min, and the highest MAE value is at a TG of 30 min. The average MAE for TF prediction with TG of 10, 15, and 30 min is 3.542, 5.006, and 5.846, respectively. In TF prediction, different time granularities have a significant impact on the accuracy of the model. When the TG is 10 min, the model can more finely capture the frequent fluctuations and short-term changes of TF, with the minimum MAE. As the TG increases to 15 and 30 min, the time step of the model becomes longer; the time resolution of the data decreases; the details of short-term changes are smoothed or lost, leading to a gradual increase in prediction error. The detailed observation ability with a 10-min TG gives it an advantage in capturing short-term dynamic changes in TF, resulting in lower MAE values. When the TG is 15 min, the balance between data volume and time resolution is achieved, followed by prediction accuracy. When the TG is 30 min, the prediction accuracy is

Table 5 Prediction performance under different windows

Time granularity (min)	Index	Window size: 30 min	Window size: 60 min	Window size: 120 min	Window size: 240 min
10	RMSE	4.478	4.565	4.671	4.773
	MAE	3.609	3.789	3.799	3.921
	R^2	0.965	0.943	0.941	0.933
15	RMSE	6.678	6.342	6.789	6.798
	MAE	5.687	5.234	5.459	5.711
	R^2	0.932	0.946	0.934	0.929
30	RMSE	9.567	9.126	8.789	7.678
	MAE	7.276	6.976	6.715	6.435
	R^2	0.921	0.926	0.931	0.938

Fig. 2 TF prediction results at different time granularities



the worst due to insufficient capture of short-term changes. Smaller TG provides higher time resolution, allowing the model to perform better in capturing traffic changes in detail, resulting in the lowest MAE value. Therefore, in subsequent research, this article selects a TG of 10 min and a window size of 30 min.

4.3 Predictive performance during different TF periods

There are significant differences in TF at different times of the day, with peak periods set from 10:00 to 19:00 and off-peak periods set for the rest of the day. A TG of 10 min and a window size of 30 min are adopted. The TF prediction results of the LSTM-BiGRU-Attention model during peak and off-peak periods are shown in Fig. 3.

The stability of the LSTM-BiGRU-Attention model is determined by comparing the MAE values of TF predictions during peak and off-peak periods. On October 1st and October 3rd to October 7th, the difference in MAE for TF predictions during peak and off-peak periods is very small, with absolute values of MAE differences being less than 0.1. On October 2nd, the MAE for peak TF prediction is 3.445, while the MAE for off-peak TF prediction is 3.345. The LSTM-BiGRU-Attention model has very similar TF prediction errors during peak and off-peak periods, indicating that LSTM-BiGRU-Attention has high stability. The LSTM-BiGRU-Attention model, by combining LSTM to capture long-term dependencies, BiGRU to capture bidirectional information, and the Attention mechanism to dynamically allocate weights, can effectively cope with dynamic changes in TF and provide stable and reliable prediction results.

4.4 Comparison of predictive performance of different models

The comparison of the predictive performance of different models is shown in Fig. 4.

The horizontal axis in Fig. 4 shows 9 compared models, with RMSE and MAE values on the left vertical axis and determination coefficient R^2 on the right vertical axis. It can be seen that the RMSE and MAE of LSTM-BiGRU-Attention are the lowest, while R^2 is the highest. The RMSE, MAE, and R^2 of LSTM-BiGRU-Attention are 4.478, 3.609, and 0.965, respectively. LSTM-BiGRU-Attention combines multiple advantages of LSTM, BiGRU, and Attention mechanisms. The excellent performance of the LSTM-BiGRU-Attention model is due to the combination of its multiple mechanisms. LSTM captures long-term dependencies, BiGRU enhances the understanding of contextual information, and the Attention mechanism improves the model's adaptability to complex traffic patterns by adaptively focusing on the impact of

Fig. 3 TF prediction results during peak and off-peak periods

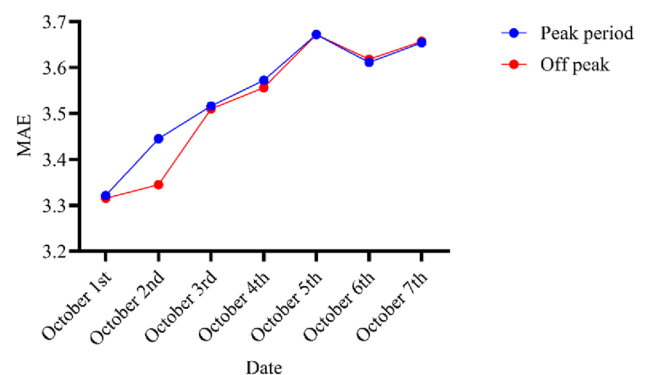
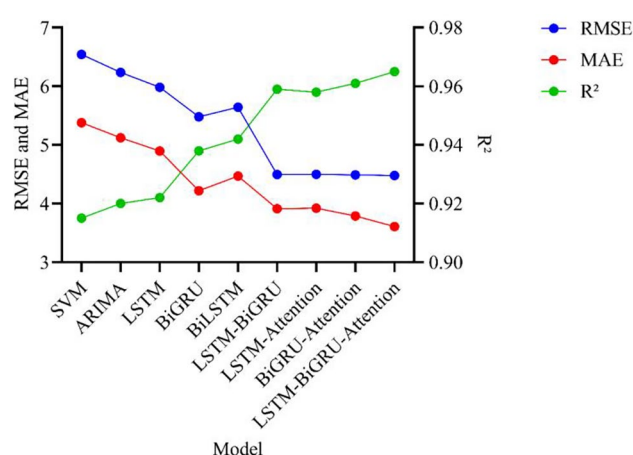


Fig. 4 Comparison of the predictive performance of different models



important moments. This enables the model to provide higher prediction accuracy when faced with nonlinear and long-period traffic flow data. In contrast, other models such as traditional LSTM or BiGRU may not be able to fully capture the complexity and time dependency of traffic flow, resulting in poor prediction results.

4.5 Model training duration and parameter quantity

The comparison of training time and parameter quantity for different models is shown in Fig. 5.

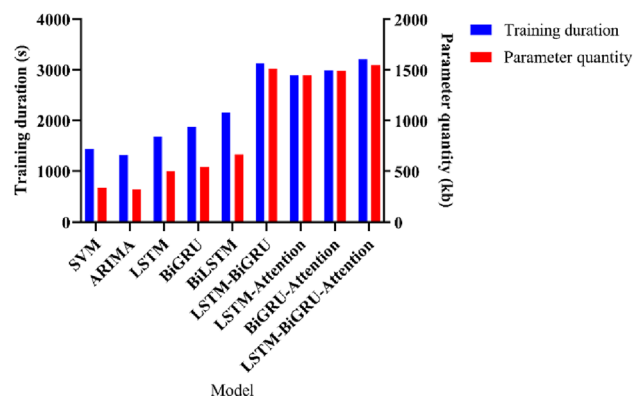
Models like the LSTM-BiGRU-Attention architecture are considerably more demanding in terms of computational resources due to their sophisticated design, combining multiple advanced mechanisms. The LSTM-BiGRU-Attention model has longer training times and more parameters than simpler models. However, its ability to identify complex patterns in time series data improves forecast consistency and accuracy. This makes it ideal for challenging forecasting tasks that require understanding and predicting intricate data correlations.

5 Application section

Real-time applications of traffic flow prediction:

The real-time application of the LSTM-BiGRU-Attention model proposed in this article in traffic flow prediction has significant practical significance. In actual traffic management, accurate flow prediction is crucial for optimizing traffic signal control, scheduling public transportation, and reducing congestion. By utilizing the LSTM-BiGRU-Attention model, the traffic management system can predict traffic flow changes in the short term in real time and provide timely decision support for the traffic control center. Through in-depth learning and processing of historical traffic data, the model can accurately capture the dynamic changes in traffic flow and make predictions based on real-time data. This real-time application can not only improve the prediction accuracy of traffic flow but also provide a scientific basis for the adjustment of traffic lights, thereby effectively reducing traffic congestion and improving traffic efficiency.

Fig. 5 Training duration and parameter quantity of different models



Public transportation scheduling optimization:

Scheduling optimization of public transportation systems is an important step in improving the efficiency of urban transportation systems. Traditional dispatching strategies often rely on static historical data and empirical rules, making it difficult to cope with complex and changeable traffic flow conditions. The LSTM-BiGRU-Attention model in this article can predict future traffic flow in real time and accurately estimate public transportation demand through in-depth analysis of traffic flow data. Based on these prediction results, the traffic management department can more flexibly adjust bus frequency, optimize route planning, and even add temporary buses during peak periods. This will not only improve the service level of the public transportation system but also effectively alleviate road congestion and increase the utilization rate of public transportation.

Integration of intelligent transportation systems:

The development of intelligent transportation systems relies on efficient data processing and analysis capabilities. The model proposed in this paper can be integrated with existing intelligent transportation systems to support smarter traffic management and control by providing accurate traffic flow prediction data. Specifically, the prediction results output by the model can be used in components such as traffic flow monitoring systems, traffic flow counters, and traffic signal control systems to enhance the intelligence level of these systems. For example, based on the prediction results, the intelligent traffic signal control system can dynamically adjust the switching cycle of the signal lights to optimize traffic flow and reduce congestion. In addition, the integration of the model can also support intelligent navigation systems, provide drivers with optimal driving routes based on real-time traffic prediction data, and improve overall traffic efficiency.

Urban transportation planning and policy formulation:

Urban transportation planning and policy development are key to ensuring the sustainable development of urban transportation systems. The LSTM-BiGRU-Attention model in this article provides accurate traffic flow prediction data and can provide a scientific decision-making basis for transportation planners and policymakers. By analyzing forecast results, planners can identify potential problem areas for traffic flow and develop targeted traffic improvements. For example, in high-traffic areas, policymakers can plan new roads, optimize existing road layouts, or add transportation facilities to meet future transportation needs. The model's predictive capabilities can also support long-term transportation planning strategies and help formulate more scientific urban transportation policies by simulating traffic flow changes after the implementation of different policies. This data-driven planning approach helps improve the overall efficiency and sustainability of urban transportation systems and improves citizens' travel experience.

6 Discussion

In the field of traffic flow prediction, many studies have applied a variety of deep learning algorithms, such as RNN, CNN, and more advanced hybrid models. These studies show the advantages of deep learning in capturing complex traffic flow patterns, especially the superiority of LSTM in processing time series data. Traditional algorithms such as RNN are weak in the long-term dependency problem, while LSTM solves this problem through its unique memory unit structure, which can better retain historical information and enhance prediction accuracy. However, many existing studies have not fully considered the noise problem in traffic flow data, and have limitations in feature selection and model optimization. Based on the improvement of the LSTM algorithm, this paper proposes a model that combines multi-feature input and noise processing mechanisms to improve the performance of traffic flow prediction. Experimental results show that compared with the traditional LSTM model, the improved model shows higher prediction accuracy and stability in various traffic scenarios, especially when the traffic flow fluctuates violently during peak hours, the improved model can more effectively cope with complex changes. In addition, by introducing a dynamic feature weight adjustment mechanism, the model in this paper further optimizes the generalization ability of the model, providing new ideas for the intelligent application of traffic flow prediction in the future.

The LSTM-BiGRU-Attention hybrid model proposed in this paper has significant innovation and advantages in traffic flow prediction. The model combines LSTM, BiGRU and Attention mechanisms, and each part is optimized for specific challenges in traffic flow prediction. LSTM networks can effectively capture long-term dependencies in time series, and are particularly suitable for modeling long-term trends and periodic changes in traffic flow; BiGRU, through its bidirectional structure, considers both forward and reverse time dependencies, enhances the comprehensive understanding of traffic flow changes, and thus improves the accuracy and adaptability of the model; the Attention mechanism gives the model the ability to dynamically weight, so that it can automatically focus on the key time steps that affect the results during

the prediction process, avoiding important information that may be ignored in traditional methods. This multi-level information extraction and weighting mechanism enables the LSTM-BiGRU-Attention model to perform well in the fast-changing and complex patterns of traffic flow, and can effectively predict traffic flow changes at different time granularities. In contrast, a single LSTM or BiGRU model may have limitations in processing complex, nonlinear and long-term dependent traffic data, and fails to fully explore the multi-dimensional information in traffic flow data, thus affecting the prediction accuracy. In addition, the innovation of this model is also reflected in its ability to make predictions at different time scales, such as 10-min, 15-min, and 30-min granularity predictions, providing a more flexible analysis method that can adapt to different application scenarios and data requirements. The model is also highly adaptable to cities with different traffic dynamics or data collection infrastructure.

Although the model performs well in Beijing's National Day Golden Week data, its method does not rely on specific geographical areas or holidays and can adapt to traffic flow data in other cities. In terms of data collection, cities can adjust according to their own traffic monitoring equipment and ensure data quality through data preprocessing, missing value filling, and outlier processing. Therefore, the flexibility and wide applicability of the model make it potentially valuable for traffic flow prediction in different types of cities, especially in cities with relatively mature data collection infrastructure, and can provide accurate and real-time traffic flow prediction support.

7 Conclusions

Based on the traditional LSTM model, this paper introduces BiGRU and Attention mechanism to improve the accuracy and stability of traffic flow prediction. Experimental results show that the LSTM-BiGRU-Attention model is significantly better than the single LSTM model and other benchmark models in multiple performance indicators, specifically the smallest RMSE and MAE values and the largest R^2 value. This result proves the superiority of the model in capturing the complex change pattern, long-term dependence and periodic fluctuation of traffic flow. By combining the long-term dependency modeling ability of LSTM, the bidirectional temporal information capture ability of BiGRU and the dynamic weighting ability of Attention mechanism, the model in this paper can more accurately predict the changes of traffic flow at different time granularities, especially the difference in prediction error between peak and non-peak hours is very small, indicating that it has high stability under different traffic flow conditions. This advantage makes the model have important application value in optimizing traffic management, improving the accuracy of traffic flow prediction and alleviating traffic congestion. However, the model in this paper also has certain limitations, especially in terms of computing resources and hardware requirements. The high complexity of the model may limit its application in resource-constrained environments. Therefore, future research can explore more efficient model optimization methods to reduce its computational overhead while maintaining its predictive performance. The model in this study relies on high-performance GPUs and large amounts of memory, which may be difficult to implement in resource-constrained environments, limiting its applicability in some practical application scenarios. In addition, the experimental data only covers traffic flow conditions during the National Day Golden Week, and the applicability of the model in other time periods and different traffic environments needs further verification. Future research can further improve the universality and practicality of the model by optimizing the model structure and algorithm, reducing computational costs, and expanding data sets and application scenarios.

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Declarations

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